

Stability

Stability

- Tâtonnement price dynamics is a nonlinear differential equation

$$\frac{d\hat{p}}{dt} = \hat{z}(\hat{p}) \text{ on } \mathbb{R}_{++}^{L-1}.$$

- A solution is a function $\hat{p} : [0, \infty) \rightarrow \mathbb{R}_{++}^{L-1}$ where $\hat{p}(t)$ solves the differential equation.
- $\hat{p}^*$ is an equilibrium point if $\hat{z}(\hat{p}^*) = 0$.

$$\forall t, \hat{p}(t) = \hat{p}^*, \frac{d\hat{p}(t)}{dt} = \hat{z}(\hat{p}^*) = 0.$$

- Do solutions converge to the WE price?

Stability definitions

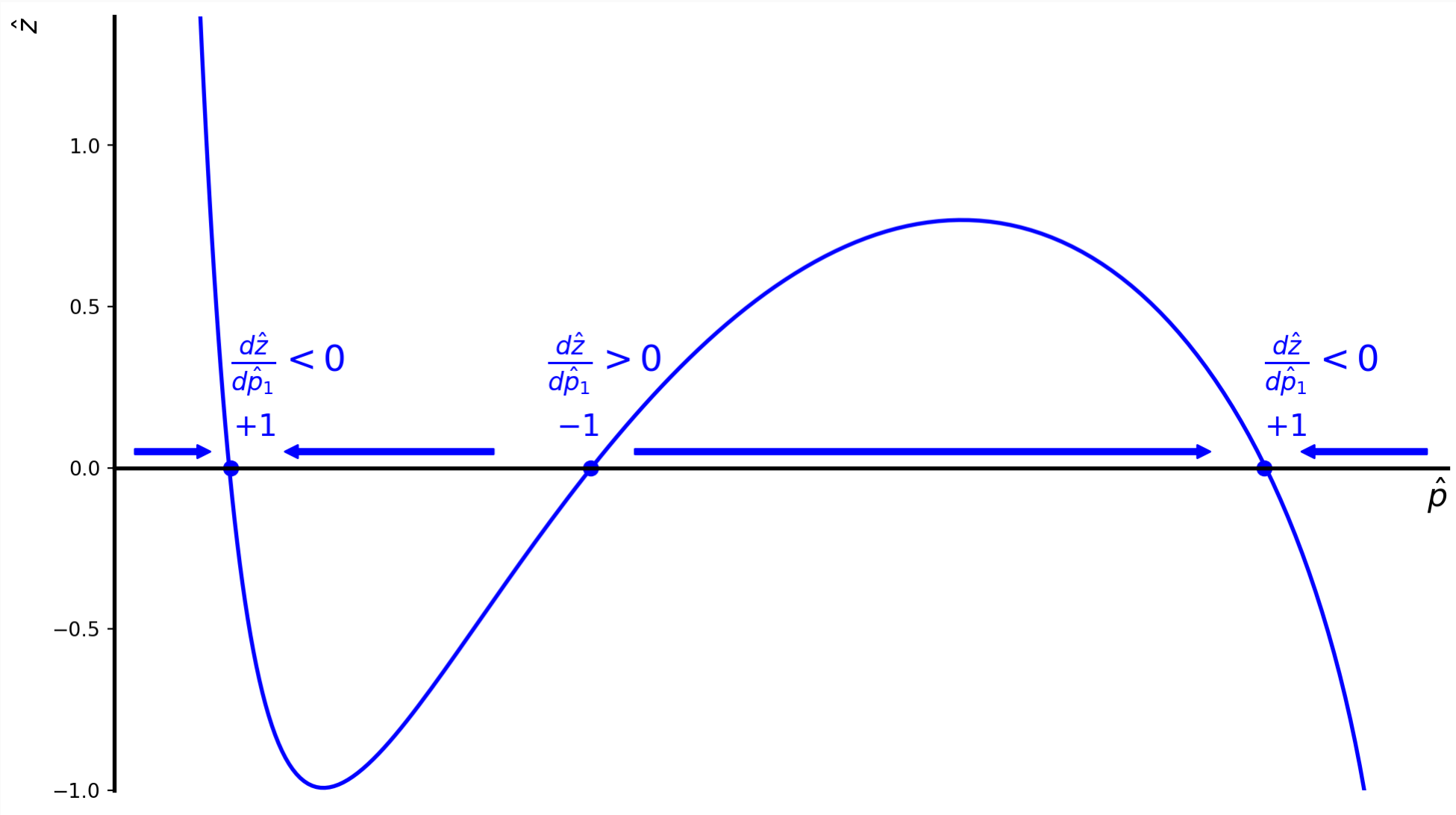
Definition 1 (Locally stable) An equilibrium point $\hat{p}^*$ is locally stable (LS) if $\forall \epsilon > 0, \exists \delta > 0 : [\|\hat{p}_0 - \hat{p}^*\| < \delta]$ implies that any solution $\hat{p}(t), \hat{p}(0) = \hat{p}_0$ satisfies $\|\hat{p}(t) - \hat{p}^*\| < \epsilon$

If $\hat{p}^*$ is not locally stable, it is unstable.

Definition 2 (Locally Asymptotically stable) An equilibrium point $\hat{p}^*$ is locally asymptotically stable (LAS) if it is locally stable and every locally stable solution $\hat{p}(t), \hat{p}(0) = \hat{p}_0$ satisfies $\lim_{t \rightarrow \infty} \hat{p}(t) = \hat{p}^*$

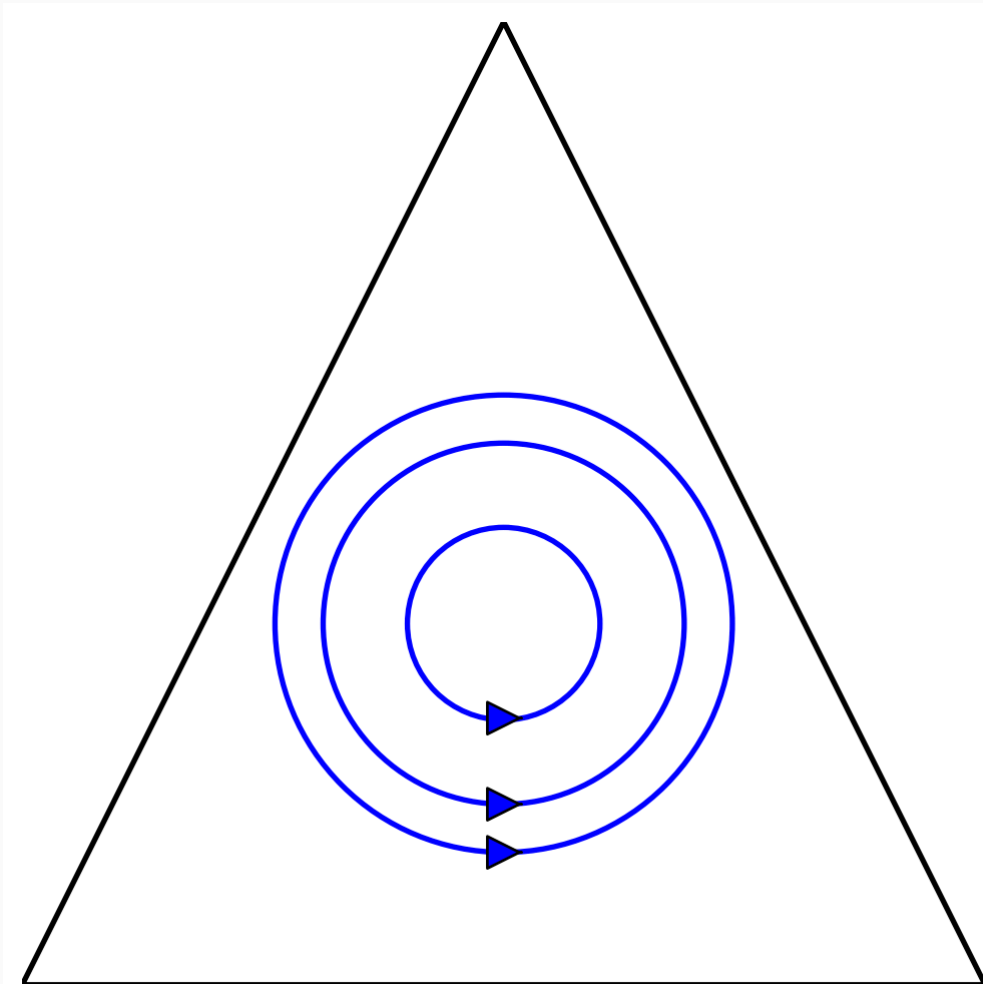
- Linear approximation: $\frac{d\hat{p}}{dt} = D\hat{z}|_{\hat{p}^*}(\hat{p} - \hat{p}^*)$ suffices to identify LAS equilibria.

Tâtonnement dynamics



Scarf example

- Scarf (1960), Good treatment in Takayama (1985), page 333.



Stability results

Theorem 1 (Hirsch and Smale) An equilibrium point $\hat{p}^*$ is locally asymptotically stable if and only if all eigenvalues of the Jacobian $D_{\hat{p}}\hat{z}$ evaluated at $\hat{p}^*$ have a negative real part.

The proof is in Hirsch and Smale (1974), chapter 9.

- If one eigenvalue has a positive real part, $\hat{p}^*$ is not LAS.

References

- Hirsch, Morris W., and Stephen Smale. 1974. *Differential Equations, Dynamical Systems, and Linear Algebra*. San Diego, Ca: Academic Press, Inc.
- Scarf, H. 1960. "Some Examples of Global Instability of the Competitive Equilibrium." *International Economic Review* 1 (3): 157–72.
- Takayama, Akira. 1985. *Mathematical Economics*. Second edition. Cambridge University Press.